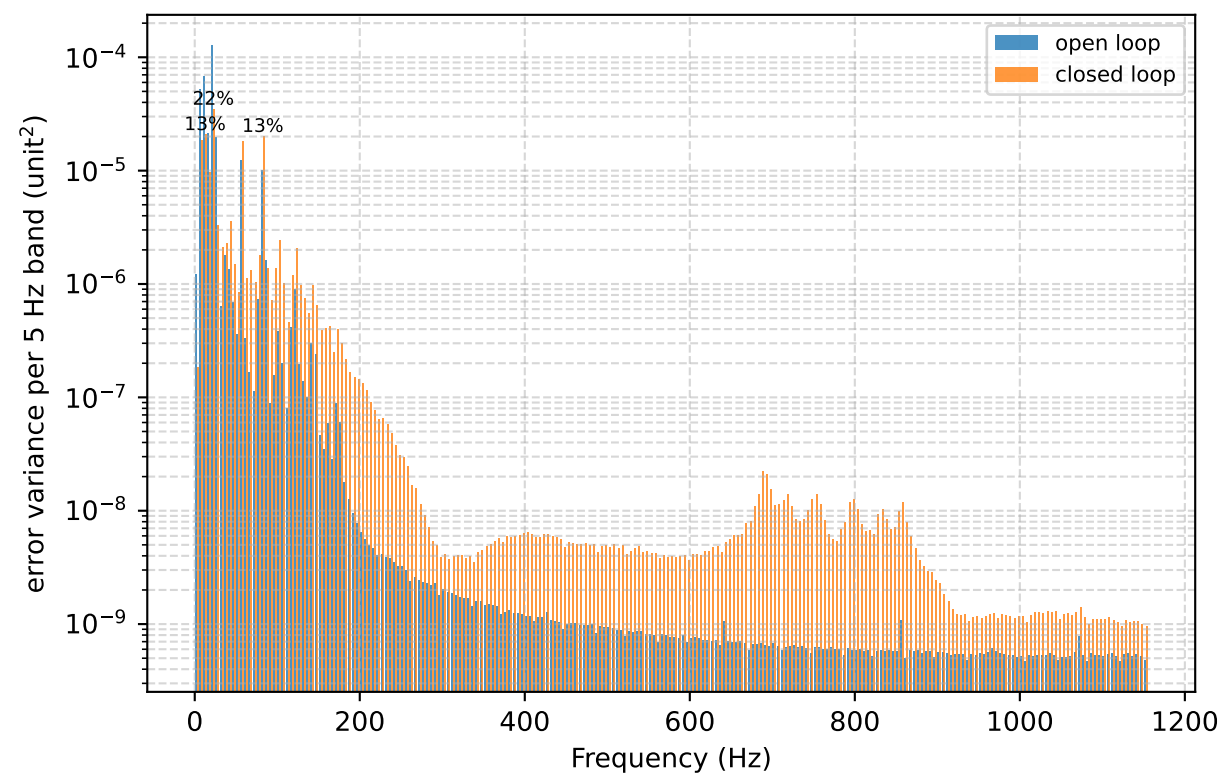
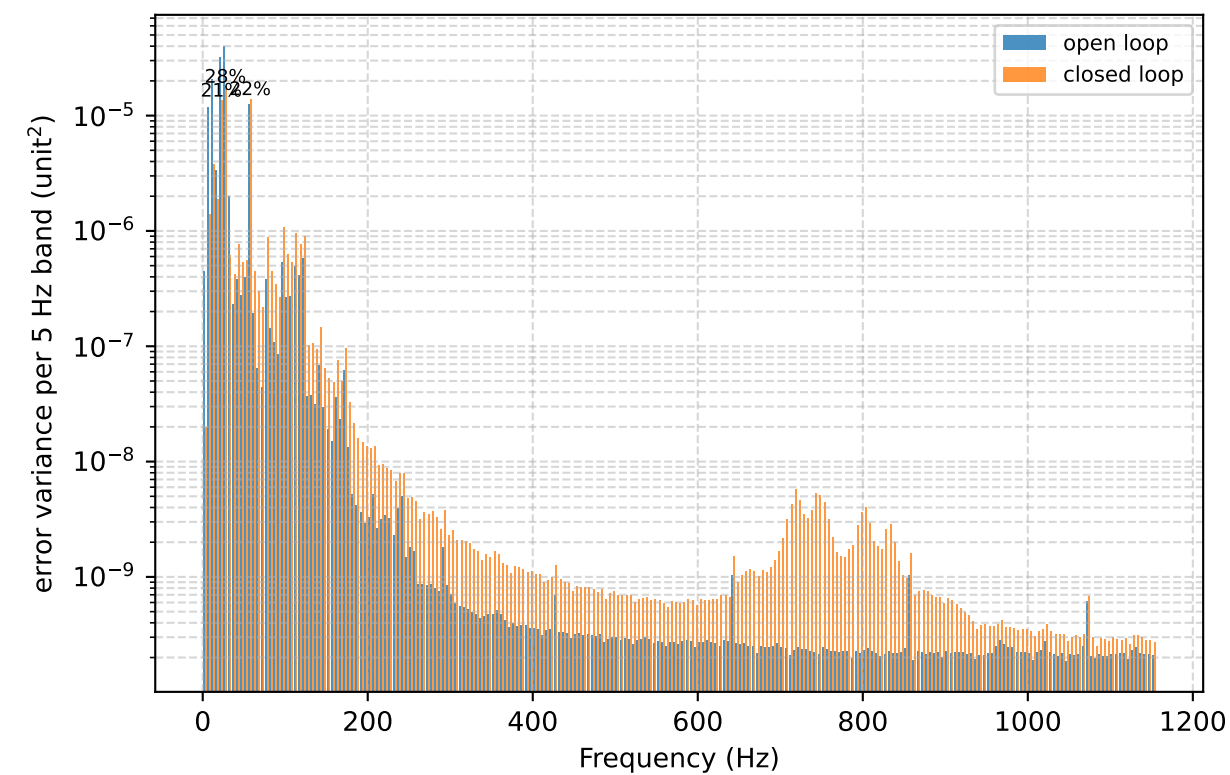
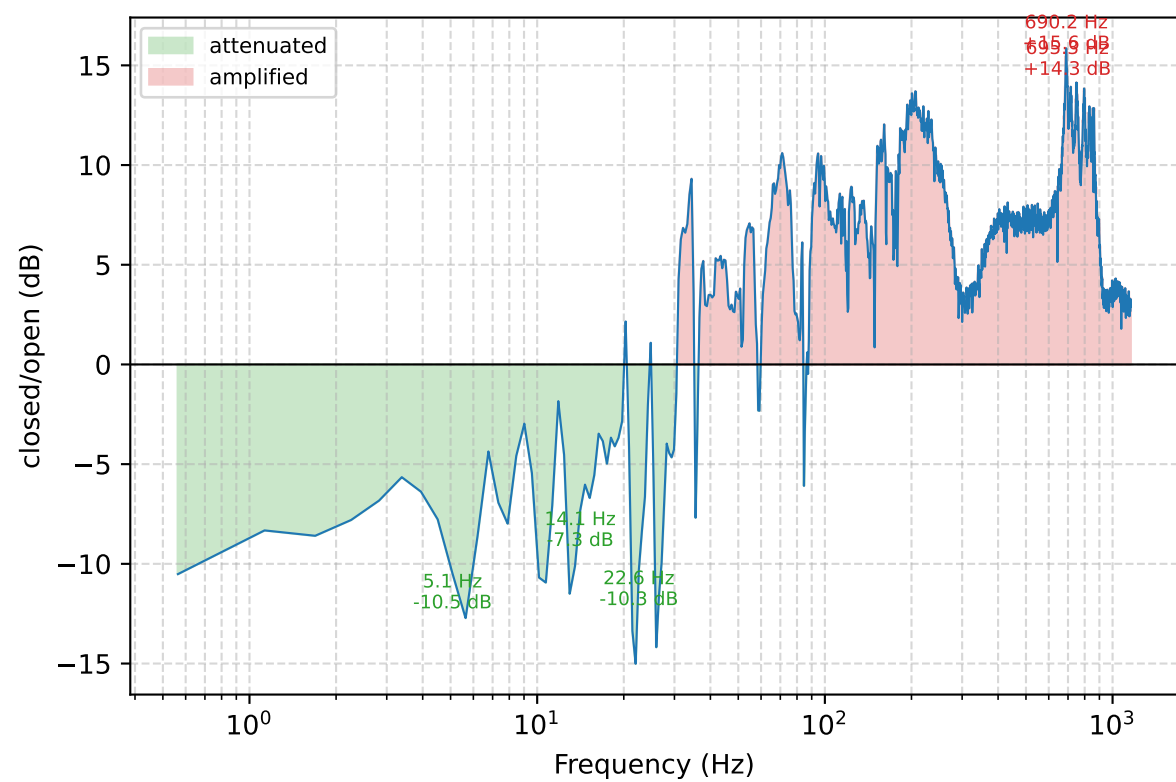
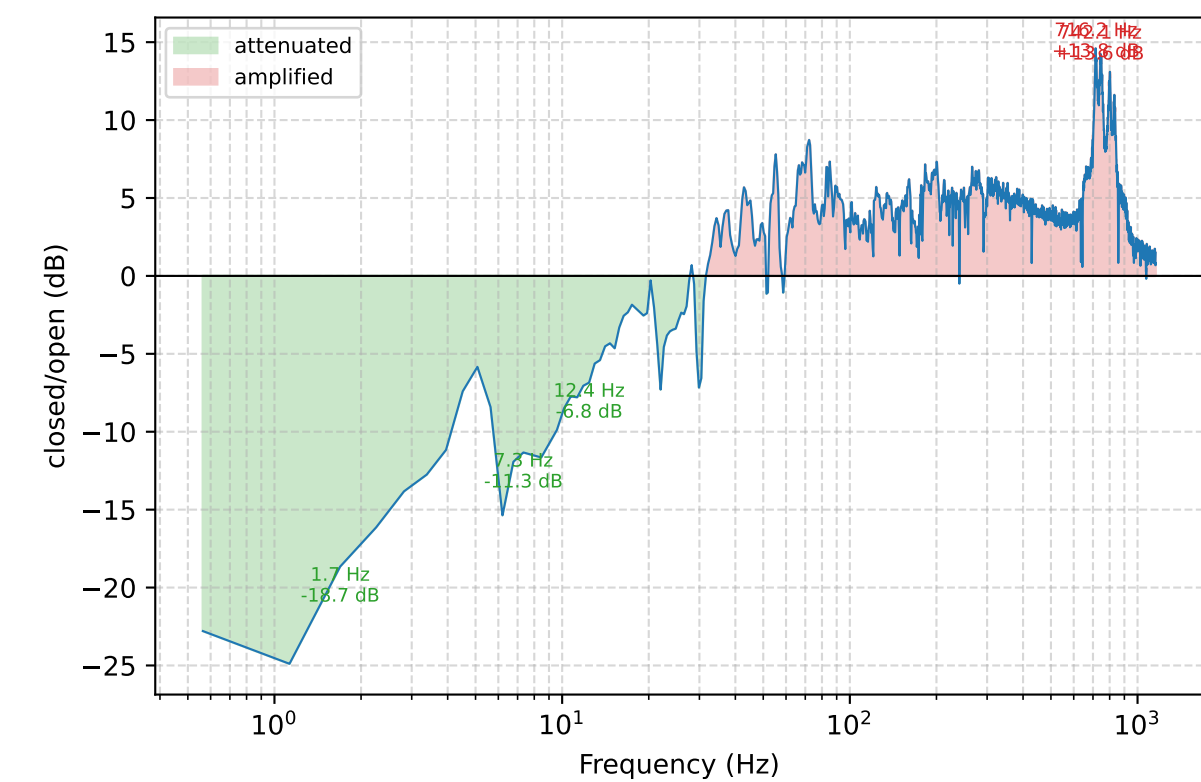
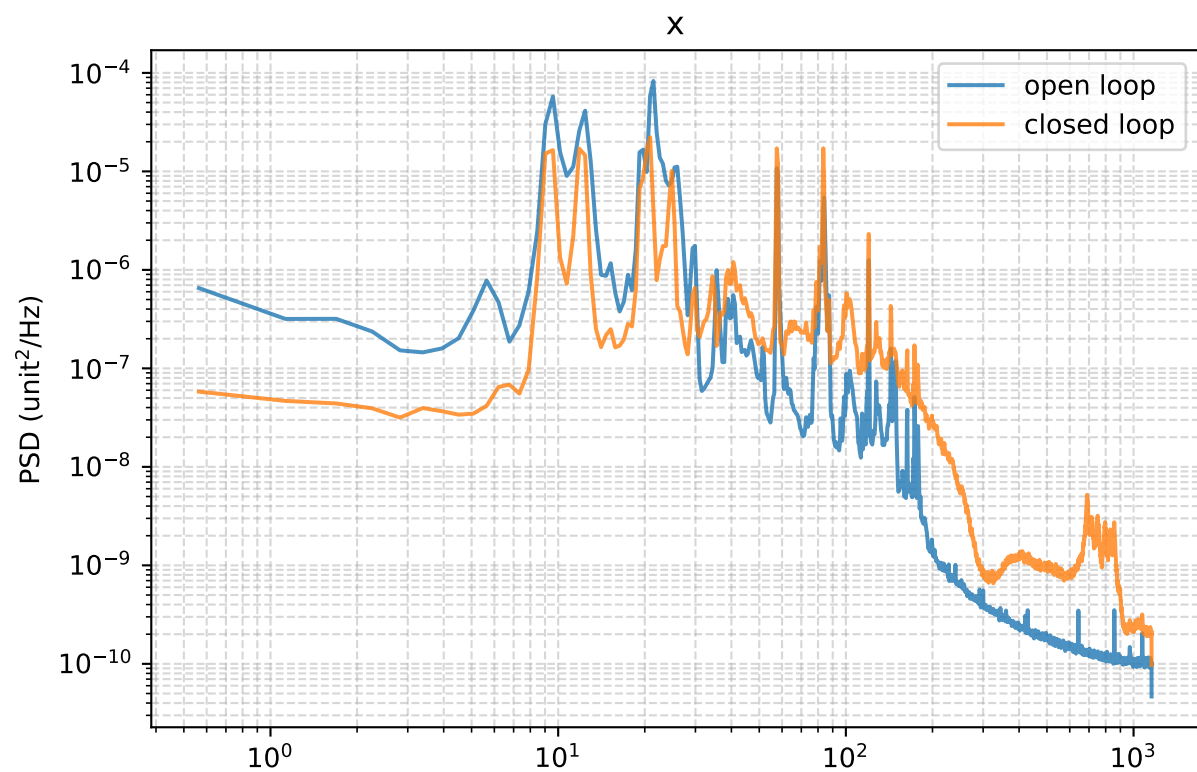
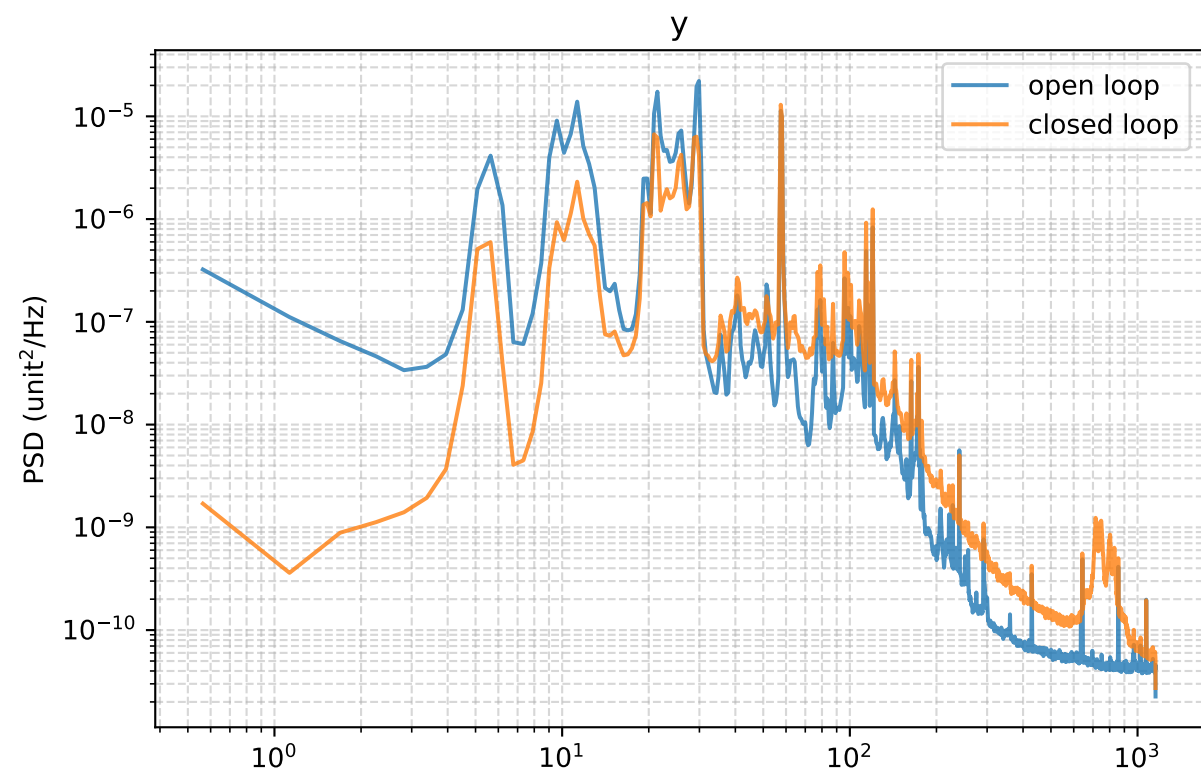


Closed-loop attenuation



Run configuration and per-band RMS

Started: 2026-07-14 10:48:04
pixel_scale: 31.5 px per output unit
y: mean +2.731 -> +0.000 px; RMS about mean 0.438 -> 0.253 px (open -> closed)
x: mean -8.281 -> -0.000 px; RMS about mean 0.590 -> 0.400 px (open -> closed)

Config used:
100, 500, 100, 500, 4096: camera ASI290MM 64x64, ROI start (920,488), ~2310 Hz loop
pid: integrator-only (p=py=d=0), i=680 iy=500, units=minmax (fine bode 2026-07-14 @ bias 2.52/1.62: Kx=0.427 Ky=0.300, tau_loop=1.81 ms; fc_x=46 Hz PM 60, fc_y=24 Hz PM 75)
lines x=[9.59,12.41,21.43,25.38,35.53,58.08,84.02] gain=[40,40,60,40,30,20,15]
linesy=[5.64,21.43,29.32,58.08] gainy=[50,20,20,15]
phases=default(-360*f*line_delay), line_delay=0.0018 (=tau loop)
kalman: order=80 horizon=5 mu=0.03 gain=1.0 gainy=0 ramp=10
piplate: fine ch0(+2)/ch1(-2), coarse bias 2.55/1.65 (bode @ 2.52/1.62), baud 460800

RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.1108	0.0373	0.2306	0.1363
10-20	0.1545	0.0749	0.2975	0.1749
20-30	0.2680	0.1765	0.3815	0.1955
30-40	0.0469	0.0322	0.0491	0.0661
40-50	0.0256	0.0360	0.0448	0.0713
50-60	0.1136	0.1200	0.1128	0.1372
60-70	0.0160	0.0272	0.0223	0.0493
70-80	0.0206	0.0331	0.0290	0.0531
80-90	0.0159	0.0280	0.1077	0.1466
90-100	0.0248	0.0365	0.0156	0.0457
100-110	0.0231	0.0339	0.0241	0.0584
110-120	0.0300	0.0413	0.0222	0.0407
120-130	0.0247	0.0316	0.0331	0.0549
130-140	0.0083	0.0141	0.0154	0.0361
140-150	0.0099	0.0145	0.0232	0.0403
150-160	0.0058	0.0100	0.0090	0.0283
160-170	0.0077	0.0112	0.0094	0.0259
170-180	0.0086	0.0113	0.0122	0.0264
180-190	0.0031	0.0061	0.0055	0.0196
190-200	0.0026	0.0053	0.0041	0.0171
200-210	0.0029	0.0052	0.0035	0.0157
210-220	0.0024	0.0043	0.0031	0.0129
220-230	0.0026	0.0041	0.0029	0.0113
230-240	0.0025	0.0038	0.0028	0.0103
240-250	0.0025	0.0036	0.0026	0.0083
250-260	0.0019	0.0031	0.0025	0.0074
260-270	0.0013	0.0026	0.0022	0.0057
270-280	0.0013	0.0027	0.0022	0.0045
280-290	0.0012	0.0024	0.0021	0.0035
290-300	0.0016	0.0025	0.0020	0.0030
total 0-300	0.3574	0.2523	0.5665	0.3989